

date 5

Topic: Fuzzy C-means (FCM) clustering

Q. what is clustering?

unsupervised learning technique where:

- we group data points
- Based on similarity
- without using class labels

Q. why fuzzy clustering?

In many real datasets:

- a point does not belong fully to only one group

- ex: i) student can be average & good
- ii) temperature can be hot & warm

so, instead of hard assignment, we use soft (fuzzy) assignment.

Q. what is fuzzy C-means?

- each data point belongs to all clusters
- with different membership values
- membership values lie between 0 & 1
- sum of memberships of a point = 1

Q. now, what does c mean & m (fuzziness parameter) means.
 c = no. of clusters

m controls how fuzzy the clustering is
always $m \geq 1$

m value

$m = 1$

$m = 2$

$m \uparrow$

meaning

becomes like k means

standard value

more overlap between clusters.

Q. what is membership matrix u ?

a matrix where?

- rows = datapoints
- columns = clusters
- entry u_{ij} = degree of belonging of point i to cluster j .

$$u := U = \begin{bmatrix} 0.8 & 0.2 \end{bmatrix}$$

means, 80% in cluster 1

20% in cluster 2.

Q. what are cluster centres?

cluster centres are weighted averages, not simple means.

they depends on:

- data values
- fuzziness parameter m .
- membership values

solving Q1

$$\text{dataset } x = \{2, 4, 6, 8\}$$

$$C=2, m=2$$

initial membership matrix given

a) objective function of FCM

FCM works by minimising this objective function:

$$J_m = \sum_{i=1}^n \sum_{j=1}^C (u_{ij})^m \|x_i - v_j\|^2$$

where it comes from:-

Step 1:- start from what k-means does

in k-means, we want to:

- assign each point to one cluster
- minimize distance from point to its cluster center

so the k-means objective function is:

$$J = \sum_{i=1}^n \sum_{j=1}^c \mu_{ij} \|x_i - v_j\|^2$$

where:

• $\mu_{ij} \in \{0, 1\}$

• if point x_i belongs to cluster j , then $\mu_{ij} = 1$

(hard assignment)

Step 2:- what change in fuzzy c-means?

FCM relaxes the hard condition

instead of: $\mu_{ij} \in \{0, 1\}$

we allow: $\mu_{ij} \in [0, 1]$ & $\sum_{j=1}^c \mu_{ij} = 1$

meaning:

a point can belong partially to multiple clusters

Step 3:- why do we have membership to power m ?

if we directly use:

$$J = \sum \mu_{ij} \|x_i - v_j\|^2$$

then:

• small membership still affect centers too much

• no control over "fuzziness"

so we introduce the fuzzy parameter m : $(\mu_{ij})^m$

∴ final objective function

$$J_m = \sum_{i=1}^n \sum_{j=1}^c (u_{ij})^m \|x_i - y_j\|^2$$

by cluster centre formula

$$y_j = \frac{\sum (u_{ij})^m x_i}{\sum (u_{ij})^m}$$

→ where does the cluster centre formula come from?
we want to find cluster centre y_j that minimize J_m .

So, we take:

• partial derivative of J_m

• w.r.t y_j

• and set it to zero.

step 1:- $J_m = \sum_{i=1}^n (u_{ij})^m \|x_i - y_j\|^2$

or

step 2:- expanding norm

$$\|x_i - y_j\|^2 = (x_i - y_j)^2$$

step 3:- Take derivative w.r.t y_j

$$\frac{\partial J_m}{\partial y_j} = \sum_{i=1}^n (u_{ij})^m \cdot 2(y_j - x_i)$$

at derivative = 0

$$\sum (u_{ij})^m (y_j - x_i) = 0$$

$$v_j \sum (w_{ij})^m = \sum (u_{ij})^m x_i$$

$$v_j = \frac{\sum (w_{ij})^m x_i}{\sum (w_{ij})^m} \rightarrow \text{Ans}$$

computing cluster centres

cluster 1 $\Rightarrow$

$$v_1 = \frac{0.8^2 \times 2 + 0.6^2 \times 4 + 0.4^2 \times 6 + 0.2^2 \times 8}{0.8^2 + 0.6^2 + 0.4^2 + 0.2^2}$$

$$= \frac{1.28 + 1.44 + 0.96 + 0.32}{1.2}$$

$$= \frac{4.0}{1.2} = 3.33$$

cluster 2 $\Rightarrow$

$$v_2 = \frac{0.2^2 \times 2 + 0.4^2 \times 4 + 0.6^2 \times 6 + 0.8^2 \times 8}{1.2}$$

$$= \frac{0.08 + 0.64 + 2.16 + 5.12}{1.2}$$

$$= \frac{8.0}{1.2} = 6.67$$

cluster 1 center ≈ 3.33

cluster 2 center ≈ 6.67

points 4 & 6 have comparable memberships $\Rightarrow$ overlap exists.

b) Two differences between k-means & fuzzy c-means
k-means

hard clustering (0 or 1)

Point belongs to only one cluster

fuzzy c-means

soft clustering (0-1)

point can belong to multiple clusters

solving Q2Dataset $\Rightarrow$

$$X = \{12, 15, 18, 22, 25, 48, 52, 55, 58, 60\}$$

given $\Rightarrow C = 2$ $m = 2$

initial membership matrix similar to 7.1.

small value $\rightarrow$ cluster 1
 large value $\rightarrow$ cluster 2

value	cluster 1	cluster 2
small value	High	Low
medium value	Balanced	Balanced
large	Low	High

taking matrix:-

Data	cluster 1	cluster 2
12	0.8	0.2
15	0.7	0.3
18	0.6	0.4
22	0.55	0.45
25	0.50	0.50
48	0.55	0.55
52	0.45	0.55
55	0.84	0.6
58	0.3	0.7
60	0.2	0.8

$$V_i = \frac{(0.8)^2 \times 12 + (0.7)^2 \times 15 + \dots}{(0.8)^2 + (0.7)^2 + \dots}$$

$$\approx 26.34$$

$$V_2 \approx 46.59$$

a) if $|u_i - u_j| < 0.01$
then the point is overlapping

from table $\Rightarrow$

22	$\rightarrow (0.5, 0.5)$
25	$\rightarrow (0.5, 0.5)$
48	$\rightarrow (0.5, 0.5)$

c) comment on cluster operation
the dataset clearly form 2 natural gaps:-
smaller values (12 to 25)
larger values (48 to 60)

the clusters are well separated, with minimal overlap,
2 fuzzy c-means effectively models uncertainty at the cluster boundaries while maintaining clear separation between the main gaps.

Implementing the code:-

1. Reading input values:-

```
int n, c;  
fin >> n;  
fin >> c;
```

2. Reading the Dataset

```
vector<double> x(n);  
for (int i=0; i<n; i++)  
    fin >> x[i];
```

• stores data point in vector x

3. Reading membership matrix

```
vector<vector<double>> U(n, vector<double>(c));  
for (int i=0; i<n; i++)
```

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```
for (int j=0; j<c; j++)  
    for (int i=0; i<n; i++)
```

step 5: fuzziness parameter

```
double m = 2.0;
```

step 6: computing cluster centers

```
vector<double> centers (c, 0.0);
```

step 7: core FCM formula implementation

```
for (int j=0; j<c; j++)  
{  
    double num = 0.0;  
    double deno = 0.0;  
  
    for (int i=0; i<n; i++) {  
        double weight = pow (U[i][j], m);  
        num += weight * x[i];  
        deno += weight;  
    }  
    centers[j] = num / deno;  
}
```

step 8: identifying overlapping data points

```
bool found = false;  
for (int i=0; i<n; i++)  
{  
    if (abs (U[i][0] - U[i][1]) < 0.1)  
    {  
        found = true;  
        break;  
    }  
}
```

Dev
5/2/24